

Umang Patel

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SUMMARY

Mechanical engineer with about two years of industrial experience, finishing an M.Eng in Mechanical Engineering at the University of Victoria. I design and build machines, from a holonomic sanitization robot to competition robots, and I work on the control behind them: constrained trajectory optimization for a 7-DOF arm and adaptive MPC. In industry I sized hydraulic systems at Bosch Rexroth and investigated equipment failures on site at a bulk terminal. Looking for roles in robotics, controls and mechanical design in Canada.

SKILLS

- **Design:** Siemens NX, NX Nastran FEA, SolidWorks (working knowledge), AutoCAD, fabrication drawings, P&IDs; MASTERCAM and CNC certified (Indo-German Tool Room)
- **Fluid systems:** hydraulic circuit design, pump, motor and valve sizing from pressure, flow and head-loss calculations
- **Control:** MPC, trajectory optimization (DDP, SQP, augmented Lagrangian), recursive least squares identification, MATLAB/Simulink, CasADi, MuJoCo, Pinocchio
- **Software and robotics:** Python, MATLAB, embedded C (ATmega2560), C++ (developing), ROS 2, NVIDIA Jetson, Git, Linux

EXPERIENCE

Trigon Pacific Terminals

Prince Rupert, BC

Mechanical Engineering Co-op

Sep 2024 – Mar 2025

- Investigated failures on stacker reclaimers, dumpers and conveyors on site under lockout/tagout, and wrote technical reports with root causes and fixes.
- Designed a conveyor washdown system end to end: 3D CAD, layout, piping, material selection, supplier coordination and AutoCAD drawings for fabrication and install.
- Produced fender support repair design packages, and led a drone-based volumetric survey proposal from vendor evaluation to business case.

Bosch Rexroth (India)

Ahmedabad, India

Apprentice Engineer, Engineering Application Department

2021 – 2022

- Designed hydraulic power systems for stackers, reclaimers, wagon tippers and paddle feeders; sized pumps, motors, valves and heat exchangers from calculations, datasheets and standards.
- Drafted circuits, manifold blocks and P&IDs in AutoCAD, kept BOMs and costing in SAP, and supported commissioning and troubleshooting of delivered systems.

University of Victoria

Victoria, BC

Teaching Assistant, MECH 242 Dynamics and MECH 430 Robotics

4 terms

PROJECTS

Medical and Sanitization Robot, B.Eng Capstone | mechanical design lead, 4-person team

2020 – 2021

- Derived the inverse kinematics of a three-wheel holonomic base and sized its motors (about 15 kg-cm needed per wheel against a 22 kg-cm stall rating).
- Compared Al 6061, SS 220 and SS 410 for the frame in NX Nastran under a 500 N load and chose SS 220; designed the spray circuit (12 V diaphragm pump, pressure and flow control valves, flow meter, six nozzles). Built and demonstrated; SSIP grant funded.

Conveyor Sorting Cell (MECH 458/554) | two-person team, ATmega2560, embedded C

2023

- Interrupt-driven ADC sensing at three stations and a linked-list FIFO queue to sort four materials; moved the stepper to an S-curve profile. Sorted 48 parts in 35 seconds with no errors.

Constrained Trajectory Optimization (M.Eng thesis) | Python, Pinocchio, MuJoCo, MATLAB, CasADi

- Implemented PDAL-DDP from first principles for 3-DOF and 7-DOF Franka arms, with a Schur complement KKT reduction and a variable time-step extension near singular configurations.
- Adaptive MPC with online recursive least squares identification: about 63% better tracking than the non-adaptive baseline in simulation.

F1TENTH Autonomous Racing, UVic CAV Racing Team | ROS 2, NVIDIA Jetson

2024

- Wiring, sensor mounting, Jetson setup and PID tuning on a nine-person team; team won the Champion Award at the 2024 IEEE Victoria Section AI Velocity Cup.

ROBOCON LDCE | mechanical team member

2018 – 2020

- Built holonomic and vector drive robots with pneumatic kicking and throwing mechanisms and a spring-loaded odometry encoder mount. DD ROBOCON 2019 national winners; competed at ABU ROBOCON 2019, Mongolia.

EDUCATION

M.Eng, Mechanical Engineering, University of Victoria

Jan 2023 – Sept 2026

B.Eng, Mechanical Engineering, L.D. College of Engineering (GTU), CGPA 8.68/10

2021